KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade
Anand Nagar, Krishnankoil, Srivilliputtur, Virudhunagar (Dt.) - 626126 | OMR, Chennai, Tamil Nadu
info@kalasalingam.ac.in | www.kalasalingam.ac.in

InterviewGen AI -
AI-Based Mock Interview and Performance Analyzer

Project Documentation

Project ID: PRJ086

Team ID: T086

Team ID: Byte-Bloom

Team GitHub Repository: https://github.com/ChSivamowny15/InterviewGen-AI

Live Link : https://interviewgen-ai-rtnk.onrender.com

Student Name	Registration Number	Individual GitHub Link
KANIKARAM MEDHARA POOJITHA	99230040116	github.com/KANIKARAMPOOJITHA
CHEEMALA SIVA MOWNY	99230040092	github.com/ChSivamowny15
GAJULA HEMALATHA	99230040300	github.com/99230040300
GANDHAM PRAVALLIKA	99230040302	github.com/pravalikagandham4

Project Trainer / Guide

Mr. SHIRWIN PRINCE I

Table of Contents

1.Introduction……...Section1
2. Problem Statement, Objectives & Scope…… ……....Section2
3. System Architecture and RAG Pipeline .....Section 3
4. Implementation Details and Technologies Used …....Section 4
5. Results, Testing and Performance Evaluation .....Section 5
6.Main Features……Section 6
7. Conclusion and Future Scope .....Section 7
8. References .....Section 8

1.Introduction

InterviewGen AI – AI-Based Mock Interview and Performance Analyzer is an intelligent web-based application designed to help students and job seekers improve their interview preparation using Artificial Intelligence. In today's competitive job market, candidates require strong technical knowledge, communication skills, and confidence to succeed in recruitment interviews. However, limited practice opportunities and the lack of personalized feedback make interview preparation challenging. To address these issues, InterviewGen AI provides an interactive platform for conducting mock interviews based on different job roles, interview categories, and difficulty levels. It evaluates candidate responses and provides personalized feedback to identify strengths and areas for improvement.

The system integrates **Retrieval-Augmented Generation (RAG)** to retrieve relevant candidate information and provide context-aware assistance, along with **Large Language Model (LLM)** technology through the Groq API using the GPT-OSS-120B model for intelligent response evaluation and feedback generation. It also includes features such as resume analysis, voice-based interview practice, speech recognition, webcam-based monitoring, performance analytics, and interview report generation. Developed using Python, Flask, SQLite, HTML, CSS, and JavaScript, InterviewGen AI aims to provide a user-friendly and effective environment for improving interview skills, building confidence, and preparing candidates for real-world recruitment processes.

2. Problem Statement, Objectives & Scope

2.1 Problem Statement

Many students and job seekers face difficulties in preparing for technical and HR interviews due to limited practice opportunities, lack of personalized guidance, and insufficient feedback

on their performance. Traditional interview preparation methods do not always provide real-time evaluation or identify individual weaknesses in technical knowledge, communication, and confidence. Therefore, there is a need for an intelligent and interactive platform that can simulate interview environments, analyze candidate responses, and provide personalized suggestions for improvement using Artificial Intelligence.

2.2 Objectives

The main objectives of InterviewGen AI are:

1. To develop an AI-powered web application that provides interactive mock interviews based on different job roles, interview categories, and difficulty levels.
2. To implement Retrieval-Augmented Generation (RAG) for retrieving relevant candidate information from uploaded resumes and providing personalized, context-aware interview assistance.
3. To extract and process resume content to identify candidate skills, educational qualifications, projects, and technical experience for relevant information retrieval.
4. To integrate the retrieved context with the Large Language Model (LLM) through the Groq API using GPT-OSS-120B to support intelligent interview question generation, answer evaluation, and feedback generation.
5. To provide additional features such as resume analysis, voice-based interview practice, speech recognition, and performance analytics.
6. To generate detailed interview performance reports highlighting candidates' strengths and weaknesses, helping them improve their technical knowledge, communication skills, and confidence.

2.3 Scope of the Project

InterviewGen AI focuses on providing a comprehensive interview preparation environment for students and job seekers. The system supports multiple interview categories, including Technical, HR, Behavioral, and Mixed interviews, with different difficulty levels. It offers resume analysis, AI-based answer evaluation, RAG-based context retrieval, voice interaction, webcam-based monitoring, performance analytics, and report generation. The platform is designed to provide accessible and personalized interview practice through a web-based interface. In the future, it can be enhanced with multilingual interview support, advanced speech analysis, improved retrieval mechanisms, and additional job-role-specific interview scenarios.

3. System Architecture and RAG Pipeline

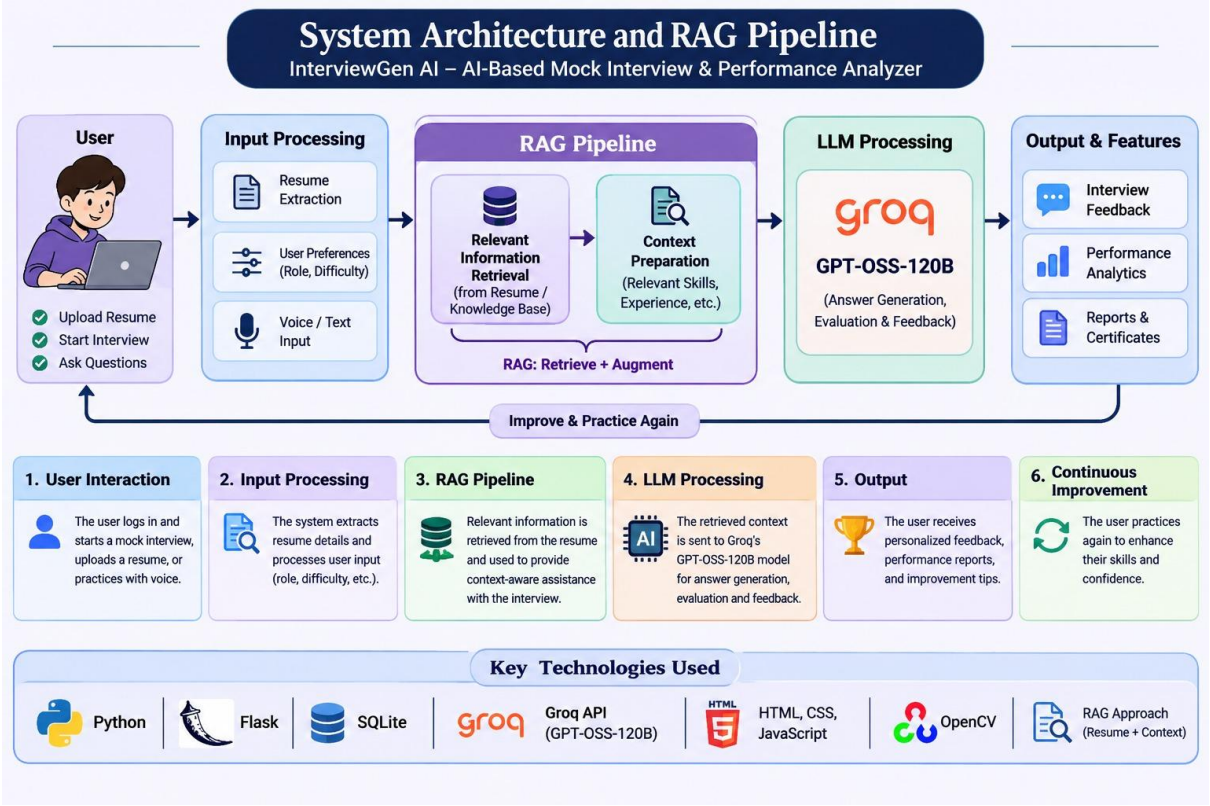


Figure 1: Overall System Architecture and RAG Pipeline of InterviewGen AI

The system architecture of InterviewGen AI consists of several interconnected modules that work together to provide an intelligent interview preparation platform. The user interacts with the application through a web-based interface developed using Flask, where resumes can be uploaded and mock interviews can be started based on selected job roles and difficulty levels. The system processes the candidate's information and uses the Retrieval-Augmented Generation (RAG) approach to identify relevant information from the resume and prepare contextual inputs for the Large Language Model (LLM).

The retrieved context is provided to the GPT-OSS-120B model through the Groq API to support interview question generation, answer evaluation, and personalized feedback. The application also provides additional functionalities such as resume analysis, voice-based practice, performance analytics, and interview report generation. The overall architecture enables candidates to practice interviews, understand their performance, identify areas for improvement, and enhance their confidence for real-world recruitment processes.

4. Implementation Details and Technologies Used

4.1 Technology Stack

The following technologies are used to develop InterviewGen AI:

Component	Technology Used	Purpose
Frontend	HTML, CSS, JavaScript	User interface, styling, and interactive features
Backend	Python, Flask	Application logic, routing, and server-side processing
Database	SQLite	Storage of user information and interview records
AI Model	GPT-OSS-120B	AI-based question generation, answer evaluation, and feedback
AI Integration	Groq API	Connecting the application with the LLM
RAG	Resume-based context retrieval	Retrieving relevant candidate information for personalized interview assistance
Resume Processing	PyPDF, python-docx	Extracting text from PDF and DOCX resumes
Image Processing	OpenCV	Webcam-based image processing
Voice Processing	Speech Recognition, Text-to-Speech	Voice-based interview interaction
Development Environment	Visual Studio Code	Application development and debugging
Deployment	Render	Hosting the web application

4.2 RAG Implementation

InterviewGen AI incorporates a Retrieval-Augmented Generation (RAG) approach to provide personalized interview assistance using candidate resume information.

The RAG workflow consists of the following stages:

1. **Document Input:** The candidate uploads a resume in PDF or DOCX format.
2. **Text Extraction:** Resume content is extracted and processed to identify relevant information.
3. **Context Retrieval:** Relevant candidate information is retrieved according to the interview context.
4. **Context Augmentation:** The retrieved information is included in the prompt provided to the language model.
5. **Response Generation:** The GPT-OSS-120B model processes the contextual input through the Groq API and generates relevant interview assistance and feedback.

4.3 Database Design

SQLite is used as the database management system to store and manage application information.

Data Entity	Information Stored	Purpose
User Information	User ID, name, email, and account details	User registration and authentication
Interview Records	Interview details, selected role, category, and difficulty	Maintaining interview history
Interview Evaluation	Candidate answers, scores, and feedback	Performance evaluation
Resume Information	Extracted resume-related information	Supporting resume analysis and personalized assistance

4.4 AI Model Integration

The application uses the Groq API to integrate the GPT-OSS-120B model. The model supports AI-powered interview question generation, candidate answer evaluation, and

feedback generation. The RAG approach provides relevant candidate context to the model, helping generate more personalized interview-related responses.

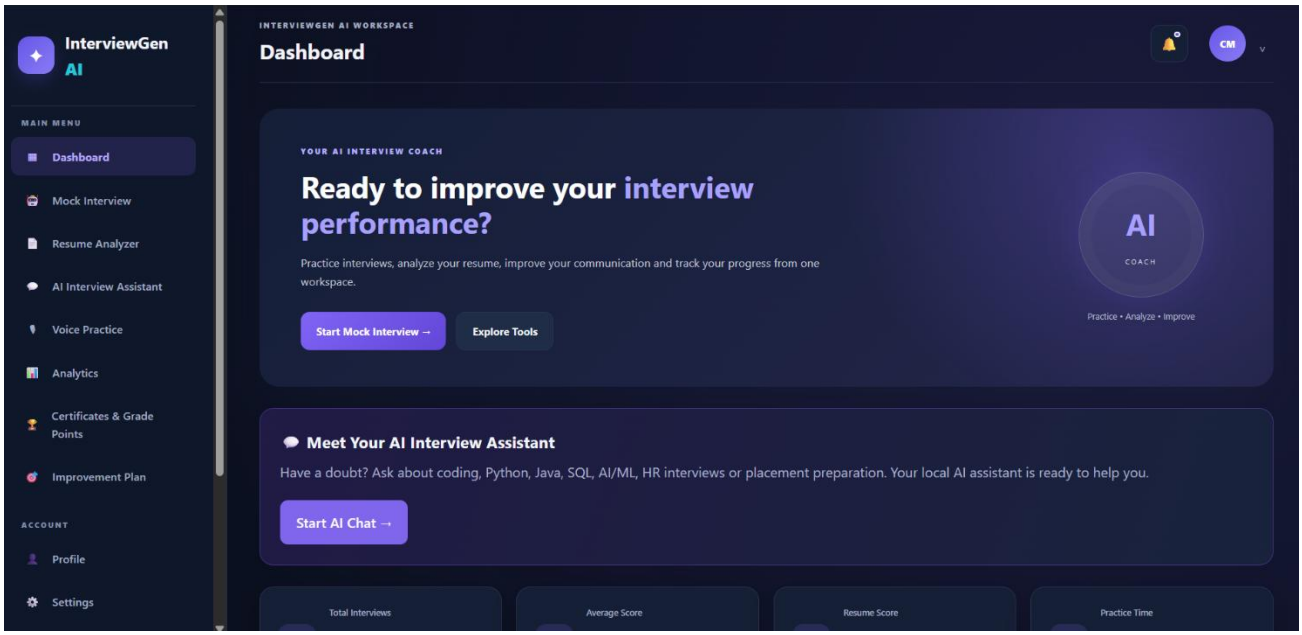
4.5 Technology Selection

Python and Flask are selected for their simplicity and flexibility in developing web applications. SQLite provides lightweight database management, while the Groq API enables integration with a large language model. The RAG approach supports candidate-specific context retrieval without relying entirely on general AI-generated information. Together, these technologies provide an integrated platform for intelligent interview preparation and performance analysis.

5.Overview of the System or System architecture

5.1 Results

InterviewGen AI was successfully developed as an AI-powered web application for interview preparation and performance analysis. The application provides an interactive platform where users can register, log in, upload resumes, and participate in mock interviews based on different job roles and difficulty levels. The RAG approach supports resume-based context retrieval, while the GPT-OSS-120B model through the Groq API provides AI-based question generation, answer evaluation, and personalized feedback. Additional features include voice-based practice, resume analysis, webcam-based monitoring, performance analytics, and report generation.



5.2 Testing

Test Module	Testing Performed	Expected Result
User Authentication	Registration and login	Successful account access
Resume Analysis	Uploading PDF and DOCX resumes	Resume information extracted
RAG Pipeline	Retrieving relevant resume information	Relevant candidate context provided to the LLM
AI Interview	Generating interview questions	Relevant questions generated
Answer Evaluation	Submitting candidate responses	Scores and feedback generated

5.3 Performance Evaluation

The performance of InterviewGen AI is evaluated based on the functionality, responsiveness, and output of its different modules. The system provides AI-generated feedback based on candidate responses and uses retrieved resume information to support personalized interview assistance. Resume analysis helps identify relevant candidate skills, while performance analytics highlight strengths and areas requiring improvement. The integrated voice and webcam functionalities provide additional support for interview practice.

5.4 Overall Outcome

The developed application demonstrates the integration of web technologies, Retrieval-Augmented Generation (RAG), and Large Language Model (LLM) capabilities in an interview preparation platform. It provides candidates with an opportunity to practice interviews, receive AI-generated feedback, analyze their performance, and improve their interview readiness through repeated practice.

6.Main Features

InterviewGen AI offers several intelligent features to support effective interview preparation and performance improvement. The major features of the application include:

1. **User Authentication and Profile Management:** Provides user registration, login, profile management, and secure access to the application.
2. **AI-Based Resume Analysis:** Extracts and analyzes relevant information from uploaded resumes, including technical skills, educational qualifications, and project details.
3. **AI-Powered Mock Interviews:** Conducts interactive mock interviews based on different job roles, interview categories, and difficulty levels.
4. **RAG-Based Context Retrieval:** Retrieves relevant candidate information from uploaded resumes and provides contextual information to the language model for personalized interview assistance.
5. **AI-Based Answer Evaluation:** Uses the GPT-OSS-120B model through the Groq API to evaluate candidate responses and generate scores, feedback, and improvement suggestions.

6. **Voice-Based Interview Practice:** Enables candidates to answer interview questions through voice input using speech recognition and text-to-speech functionalities.
7. **Webcam-Based Monitoring:** Uses OpenCV-based image processing to support webcam-related interview monitoring.
8. **Performance Analytics:** Analyzes interview performance, tracks scores, and identifies candidates' strengths and weaknesses.
9. **Interview Report Generation:** Provides interview performance reports with evaluation results and suggestions for improvement.
10. **AI Chatbot Assistance:** Offers an interactive AI assistant to support candidates with interview preparation and related queries.
11. **Certificates and Grade Points:** The dashboard includes an achievements section where users can view certificates earned and accumulated grade points. These values are expected to depend on completed interview sessions and the project's achievement rules.

7. Conclusion and Future Scope

7.1 Conclusion

InterviewGen AI demonstrates how Artificial Intelligence can be applied to make interview

preparation more effective, accessible, and personalized. The project brings together resume-based information retrieval, language model capabilities, and interactive web technologies to create an environment where candidates can practice, evaluate their responses, and learn from their mistakes. By providing structured feedback and performance insights, the system encourages continuous improvement and independent learning. It also demonstrates the practical integration of RAG and LLM technologies in an educational and career-development application. Thus, InterviewGen AI serves as a useful step towards making AI-assisted interview preparation more accessible to students and job seekers.

7.2 Future Scope

The future scope of InterviewGen AI includes:

1. Advanced RAG Implementation: Improve information retrieval using vector-based search.
2. Multilingual Support: Enable interview practice in multiple languages.
3. Advanced Speech Analysis: Evaluate pronunciation, fluency, speaking speed, and filler words.
4. Adaptive Interviews: Generate questions dynamically based on candidate performance.
5. Detailed Analytics: Provide long-term performance tracking and progress comparison.
6. Expanded Job Roles: Include more industry-specific technical and non-technical interview categories.

8.References

1. P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020. https://arxiv.org/abs/2005.11401.

2. Y. Gao, Y. Xiong, X. Gao, et al., "Retrieval-Augmented Generation for Large Language Models: A Survey," 2023. https://arxiv.org/abs/2312.10997.

3. V. Karpukhin, B. Oguz, S. Min, et al., "Dense Passage Retrieval for Open-Domain Question Answering," Proceedings of EMNLP, 2020. https://arxiv.org/abs/2004.04906.

4. A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention Is All You Need," Advances in Neural Information Processing Systems (NeurIPS), 2017. https://arxiv.org/abs/1706.03762.

5. T. B. Brown, B. Mann, N. Ryder, et al., "Language Models are Few-Shot Learners," Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020. https://arxiv.org/abs/2005.14165.

6. N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), 2019. https://arxiv.org/abs/1908.10084.